

BE A TOPPER GUIDE: MATHEMATICS

Exam-Oriented Revision Plan Based on Series EF1GH/5 & PQ1RS/1 Analysis

Created for Thunderstudy By Aashi

Priority 1: Guaranteed 5-Mark Questions (Section D)

1. Matrices: Solving Systems of Linear Equations ($AX = B$)

The Pattern: 3 equations (x, y, z) or a word problem (furniture/production). Solve using $X = A^{-1}B$.

Variation: Finding A^{-1} first and using it to solve equations.

Focus: Determinant row expansion and correct Cofactor calculation.

2. 3D Geometry: Shortest Distance & Image

The Pattern: Shortest Distance between skew lines ($r = a + \lambda b$).

Variation: Image of a point or Foot of the Perpendicular to a line.

Formula: $|((b_1 \times b_2) \cdot (a_2 - a_1)) / |b_1 \times b_2||$

3. Integrals: Area Under the Curve

Frequent Shapes: Ellipse ($x^2/a^2 + y^2/b^2 = 1$), Parabola + Line ($x^2 = 4y$), and Circle regions.

Focus: Graph sketching and limit identification.

4. Linear Programming (LPP)

Pattern: Graphically Maximize/Minimize $Z = ax + by$ subject to linear constraints.

Focus: Feasible region and corner point calculation.

Priority 2: High-Probability Case Studies (Section E - 4 Marks)

1. Probability: Bayes' Theorem & Total Probability (Defective items, medical tests, air turbulence scenarios).

2. Relations & Functions: Equivalence Relations (Reflexive, Symmetric, Transitive) often themed around students, museum, or parallel lines.

3. Application of Derivatives: Maxima/Minima (Optimization of volume, surface area, or production cost).

Priority 3: Recurring 2 & 3 Mark Topics (Sections B & C)

- **Differential Equations:** Linear Form (Integrating Factor) and Homogeneous ($y = vx$ substitution).
- **Inverse Trig:** Domain and Range of $\sin^{-1}$, $\cos^{-1}$ functions; simplification of $\tan^{-1}$ expressions.
- **Matrices:** Showing $A^2 - 5A + 7I = 0$ type polynomial proofs and finding A^{-1} through them.
- **Vectors:** Unit vector perpendicular to both (Cross Product) and Area of Parallelogram.

Summary Checklist for Revision

- ☐ **Matrix Method:** Can you solve $X = A^{-1}B$ for a 3×3 system?
- ☐ **3D Geometry:** Can you calculate the Shortest Distance between two lines?

- ☐ **3D Geometry:** Can you find the Image of a point across a line?
- ☐ **Integrals:** Can you calculate the area of an Ellipse (πab) using integration?
- ☐ **LPP:** Can you plot constraints and find Corner Points?
- ☐ **Probability:** Can you apply Bayes' Theorem to a word problem?
- ☐ **Diff Eq:** Can you solve Linear Differential Equations ($dy/dx + Py = Q$)?
- ☐ **Relations:** Can you prove a relation is Equivalence (R-S-T)?

Be A Topper Guide By Aashi Mathematics
Analyze. Revise. Dominate.